

Critical Design Review (CDR)

Architecture Version: v5.1.x **Perspective:** Principal Software Engineer / Systems Architect

1. Executive Summary

The Magnetic Field Scanner v5.1.x represents a fundamental paradigm shift from legacy polling loops to a deterministic, RTOS-driven sensor fusion engine. This document serves as the formal architectural record of the mathematical and hardware-level breakthroughs that allow the ESP32-S3 to achieve true nanotesla-level spatial gradiometry.

2. The I2C Phase Collision & The POLL Architecture

In legacy versions (v4.x.x), the RM3100 sensors were placed in Continuous Measurement Mode (CMM). The ESP32 would wait for a hardware `DRDY` (Data Ready) interrupt and blindly fetch the results.

The Problem: Sensor Lapping

Because the ESP32 handles dynamic SD Card writes and complex LVGL UI rendering, the I2C polling loop was occasionally subjected to latency spikes (e.g., waiting for an SD card block flush). While the ESP32 was blocked, the RM3100 sensors continued measuring autonomously. If a sensor finished a *second* measurement before the ESP32 fetched the *first*, the sensor would hold the `DRDY` pin permanently `HIGH` and refuse to clock out data, resulting in a permanent bus lockup.

The Solution: Deterministic POLL Mode

In v5.1.x, we completely stripped the sensors of their autonomy. The RM3100 is now operated exclusively in **POLL Mode**.

1. The ESP32 broadcasts a simultaneous `REG_POLL` command to all sensors across the I2C bus.
2. The Sensor Task drops into a 0% CPU `vTaskDelay` via FreeRTOS Event Groups, waiting for the physical `DRDY` GPIO interrupts to assert.
3. Because the measurement is explicitly triggered by the master, it is impossible for the sensor to "lap" the CPU. The phase-lock between the TIP and REF sensors is mathematically guaranteed, ensuring the spatial gradient calculation is comparing the exact same slice of time.

3. Universal Calibration & The `nT` Pre-Normalization

Magnetic sensors natively output raw integer counts (LSBs). In legacy versions, the Kabsch calibration algorithm was applied directly to these raw counts.

The Problem: Cycle Count Scaling

The RM3100's hardware gain is governed by its Cycle Count (CC) register. A reading of 1000 at $cc=200$ represents a vastly different magnetic field than 1000 at $cc=800$. Furthermore, the sensor exhibits non-linear Zero-Field Offsets (ZFO). Attempting to scale the Hard-Iron calibration matrix in software to account for CC changes resulted in massive mathematical drift and UI pinning.

The Solution: Domain Shifting

In v5.x.x, the architecture intercepts the raw LSBs and immediately converts them into physical **nanoTeslas (nT)** *before* any calibration is applied.

- By shifting the data into a physical domain, the resulting Kabsch Hard-Iron and Soft-Iron matrices become **mathematically dimensionless**.
- **The Victory:** You can now generate a highly accurate calibration ellipsoid at $cc=400$ (for high-speed UI rendering), and then drop the wand into a deep-penetration $cc=3200$ scan without changing the calibration matrix at all. The calibration is truly universal.

4. Architectural Scalability: The 3rd Sensor (NEAR)

The wand is physically designed to support a 3rd RM3100 sensor, placed 8 inches from the TIP (the NEAR sensor).

Because the v5.1.x architecture uses synchronized POLL triggering and FreeRTOS Event Groups, adding the 3rd sensor requires zero fundamental architectural changes. The ESP32 simply requires the 3rd `DRDY` interrupt bit to set in the Event Group before unblocking the I2C read cycle. This guarantees that 1st-order gradients (TIP-REF) and 2nd-order gradients (TIP-NEAR-REF) are perfectly phase-aligned without introducing any software bottlenecks.

5. Phase 11 Physics (v5.1.x): Audio Mapping & Core Saturation

The Cube-Root Audio Algorithm (`cbrtf`)

In legacy versions, mapping the calculated gradient (nT) to audio frequency (Hz) was done linearly or with aggressive exponential curves. This felt unnatural in the field (the "squashed" sensation).

- **The Physics:** The wand detects buried anomalies that act as magnetic dipoles. Dipole magnetic fields decay spatially according to the inverse-cube law ($1/r^3$).

- **The Solution:** By applying a mathematically pure cube-root (`cbrtf()`) to the measured nT gradient before scaling it to the Audio frequency bounds, the software exactly counteracts the physical $1/r^3$ decay. The resulting audio pitch ramps up linearly relative to the user's physical $1/r$ walking distance toward the target. This provides a remarkably intuitive user experience.

Saturation Failure Modes

Through rigorous physical characterization (bringing an 800-pound pull magnet directly to the tip), we proved that the PNI RM3100 exhibits two entirely distinct failure states:

1. **Digital Integer Overflow:** Occurs in high-gain states (like `cc=3200`). The raw count output simply exceeds the bounds of the internal silicon registers (maxing out around $\sim 88 \mu\text{T}$). The sensor still attempts to fire interrupts, but the data is clipped.
2. **Physical Core Collapse:** Occurs across all Cycle Counts when subjected to extreme fields (e.g., $> 533 \mu\text{T}$). The physical inductors in the RM3100 core saturate, causing the internal resonant circuit to fail. When this happens, the sensor permanently locks up and ceases firing `DRDY` interrupts until the magnetic field is removed.

Understanding this $533 \mu\text{T}$ physical ceiling is critical; it defines the absolute upper bound of the instrument's dynamic range regardless of how low we set the Cycle Counts.

The Destructive Null Dip (Return Path)

When sweeping a large linear dipole (like a 3-foot rebar rod), the reference sensor (located 24 inches back) passes through the negative return path of the magnetic flux lines. This creates a $\sim 3 \mu\text{T}$ inverted "dip" in the reference sensor data exactly as the tip sensor hits peak saturation. The software gradient math must remain aware that the reference sensor is rarely in a pure "background" state during near-field interaction.

Appendix A: Mathematical Target Modeling

5/8" Rebar Monopole Calculation

When testing the wand's detection capabilities, a standard $5/8" \times 36"$ steel property pin is used. To understand the physics of detection, we model the pin as a highly permeable ferromagnetic cylinder immersed in the Earth's background field.

Variables:

- Earth's background field (B_{earth}): $\sim 50,000 \text{ nT}$ (vertical component dominates at mid-latitudes).

- Relative Permeability of Steel (μ_r): ~100 to 1000.
- Cylinder dimensions: $r = 0.3125$ inches, $L = 36$ inches.

Because the rebar is highly permeable and vertically aligned, it acts as a flux concentrator. The Earth's magnetic field lines "funnel" into the steel, exiting at the top tip, creating a localized magnetic monopole.

The induced magnetic moment (m) of a slender rod in a uniform field is roughly proportional to its volume and the external field: $m \approx \frac{\mu_r - 1}{N} \cdot V \cdot B_{\text{earth}}$ (Where N is the demagnetization factor, heavily dependent on the length/diameter ratio).

At the surface (e.g., 1 inch above the tip), this flux concentration easily reaches a localized magnitude of **~50,000 to 500,000 nT** depending on the specific alloy's permeability.

However, because the sensor is a spatial gradiometer taking the difference between two points, the detected anomaly magnitude decays rapidly. While a true dipole decays at $1/r^3$, the near-field gradient of a monopole concentrates heavily at the source. To achieve a Signal-to-Noise Ratio (SNR) of 3:1 against the wand's empirical $\pm 0.33 \mu\text{T}$ noise floor, the physical gradient must be $\approx 1.0 \mu\text{T}$. Based on inverse-cube spatial decay, this threshold is crossed at approximately **18 to 24 inches** of depth.